C++ DirectX 11 RPG

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1

1:

	Isometric Client — DX11 RPG C++, DirectX 11, Protobuf 3, Dear ImGui, WinSock2 ServerCore v4 GameServer (WSL2 Linux) Binary Header `[size:uint16][msgId:uint16]` + Protobuf 3 payload , , , C++ 6,200 , HLSL 3 (316), Protobuf 7 (59), 26

DirectX 11 API RPG . ServerCore v4 , ServerCore .

```
isometric_client/
  src/
     Core/       App, DX11Device, EngineContext, Input, Timer, WinMain
     Game/       PlayerController, EntityManager, CombatManager,
                  DungeonGenerator, SessionState
     Renderer/   Pipeline, Camera, MeshCache, MaterialManager, ObjLoader,
                  InstanceRenderer, EffectRenderer, MinimapRenderer,
                  TextureLoader, Material
     Network/    TcpClient, PacketFramer, PacketHandler, PacketBuilder,
                  RecvBuffer, NetworkContext
     Scene/      IScene, LoginScene, CharSelectScene, LobbyScene, GameScene
     UI/         UIManager
     Data/       PlayerData, SkillData, InventoryData, CurrencyData, ChatHistory
  assets/
     models/     29 OBJ + MTL (Kenney     +   )
     shaders/    default.hlsl, effect.hlsl, minimap.hlsl
     textures/     .
  proto/          7 Protobuf   (59   )
```

2

2.1

Scene `App&` . Scene DX11Device , , Scene App . .

`App&` Context (`EngineContext`, `SessionState`, `NetworkContext`) . Scene , / / .

2: Context

EngineContext	DX11Device, Pipeline, Camera, MeshCache, MaterialManager, Input, UIManager, assetsDir, pipelineOk	
SessionState	PlayerData, InventoryData, CurrencyData, ChatHistory, playerName, playerId, isLocalPlayerDead, MapData , pendingEnterGame	+ Proto
NetworkContext	TcpClient, PacketHandler, PacketBuilder, BufferedPacket , sceneReady + Connect/Disconnect/FlushBuffered API	+

`SessionState` Proto (`FillLocalPlayerFromProto`) (`Reset`) . `Reset()` (Inventory, Currency, ChatHistory) .

`NetworkContext` `Connect(host, port)` / `Disconnect()` / `IsConnected()` API .

Scene SceneManager .

1 App.cpp --- Scene

```
sceneManager_.Register(SceneId::Game, [this]() {
    return std::make_unique<GameScene>(engineCtx_, session_, netCtx_, sceneManager_);
});
```

Game/ 4 (PlayerController, EntityManager, CombatManager, DungeonGenerator) App , GameScene mediator .

2.2

`WinMain` . `WndProc` `WM_SOCKET` `TcpClient::PollRecv()` , .

`Input.NewFrame → PeekMessage/DispatchMessage(WM_SOCKET ) → Timer.Tick(dt) → App.Update(dt) → App.Re`

`App::Update(dt)` `SceneManager::Update(dt)` , `App::Render()` `BeginFrame() → Scene.OnRender() → UIManager.BeginFrame() → Scene.OnUI() → UIManager.EndFrame() → EndFrame()` .

2.3

`SceneManager::ChangeScene()` , `hasPending_` `Update()` . `ChangeScene()` , Scene Scene (reentrancy) . Scene .

3:

LoginScene	·	C/S_LOGIN, C/S_REGISTER
CharSelectScene	· ·	C/S_CHAR_LIST, C/S_CREATE_CHAR, C/S_SELECT_CHAR
LobbyScene	· ·	C/S_ROOM_LIST, C/S_CREATE_ROOM, C/S_JOIN_ROOM

GameScene	(26) · · · · ·

2.3.1

.

. `CharSelectScene` `S_SELECT_CHAR`, `S_INVENTORY_INIT`, `S_CURRENCY_INIT` , TCP recv . LobbyScene . `gotSelectChar_`, `gotInventoryInit_`, `gotCurrencyInit_` true `TryFinishSelection()` .

. `S_JOIN_ROOM` `S_PLAYER_LIST`, `S_SPAWN`, `S_SKILL_DATA` . (LobbyScene → GameScene) , .

`NetworkContext` 200~399 `buffered` . `GameScene::OnEnter()` 26 `sceneReady = true` `FlushBuffered()` .

2 `App.cpp --- NetworkContext`

```
framer_.SetCallback([this](uint16_t msgId, const uint8_t* payload, int payloadLen) {
    if (!netCtx_.sceneReady && msgId >= 200 && msgId <= 399) {
        netCtx_.buffered.push_back({ msgId, std::vector<uint8_t>(payload, payload + payloadLen) });
        return;
    }
    packetHandler_.Dispatch(msgId, payload, payloadLen);
});
```

3 `GameScene.cpp ---`

```
net_.sceneReady = true;
net_.FlushBuffered();  //          Dispatch
```

3

3.1

`Pipeline` `default.hlsl` VS/PS , 2 Constant Buffer .

4:

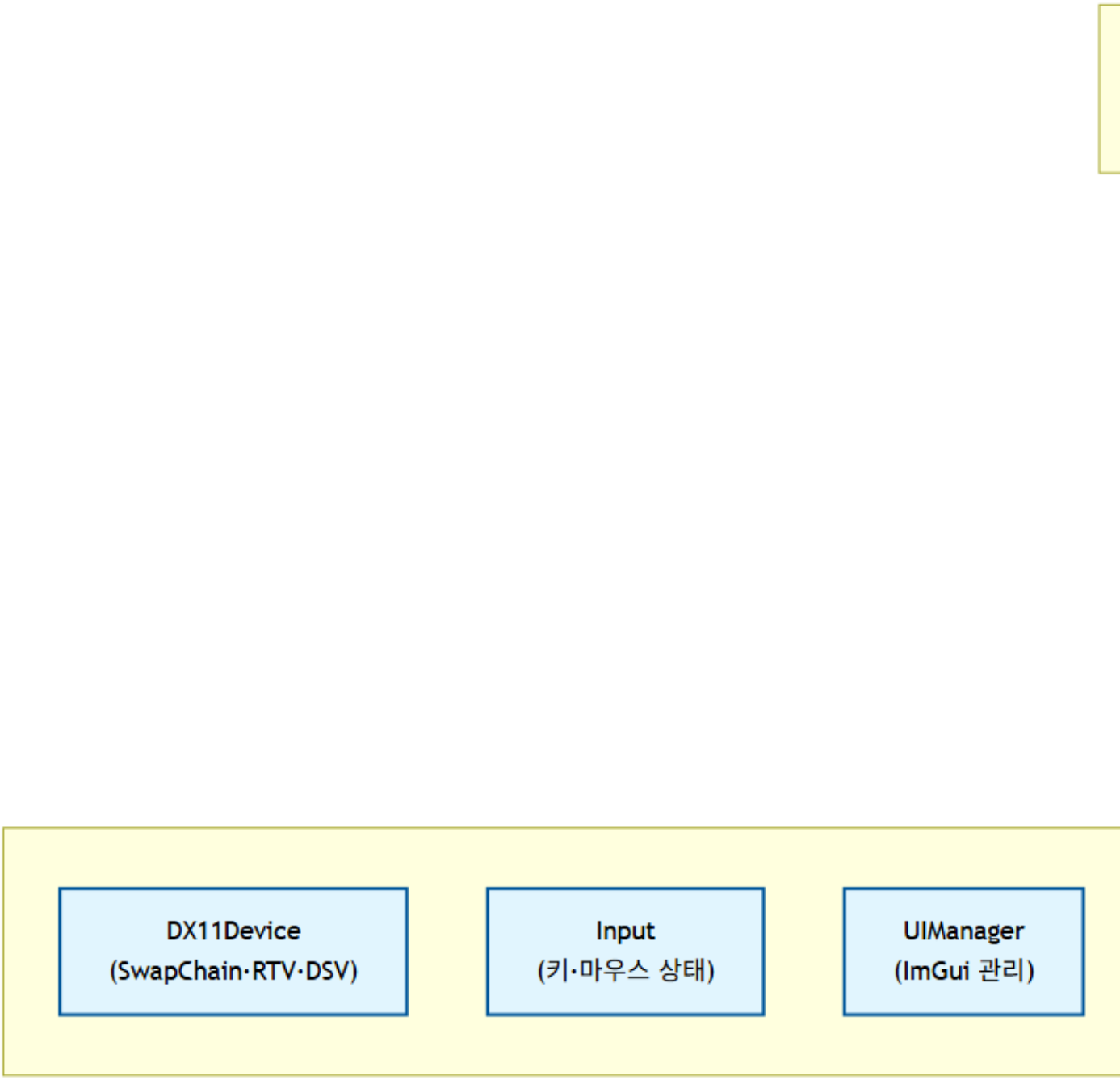
PerFrame	`b0`	ViewProj , LightDir, AmbientColor
PerObject	`b1`	World , HitFlash (0~1)

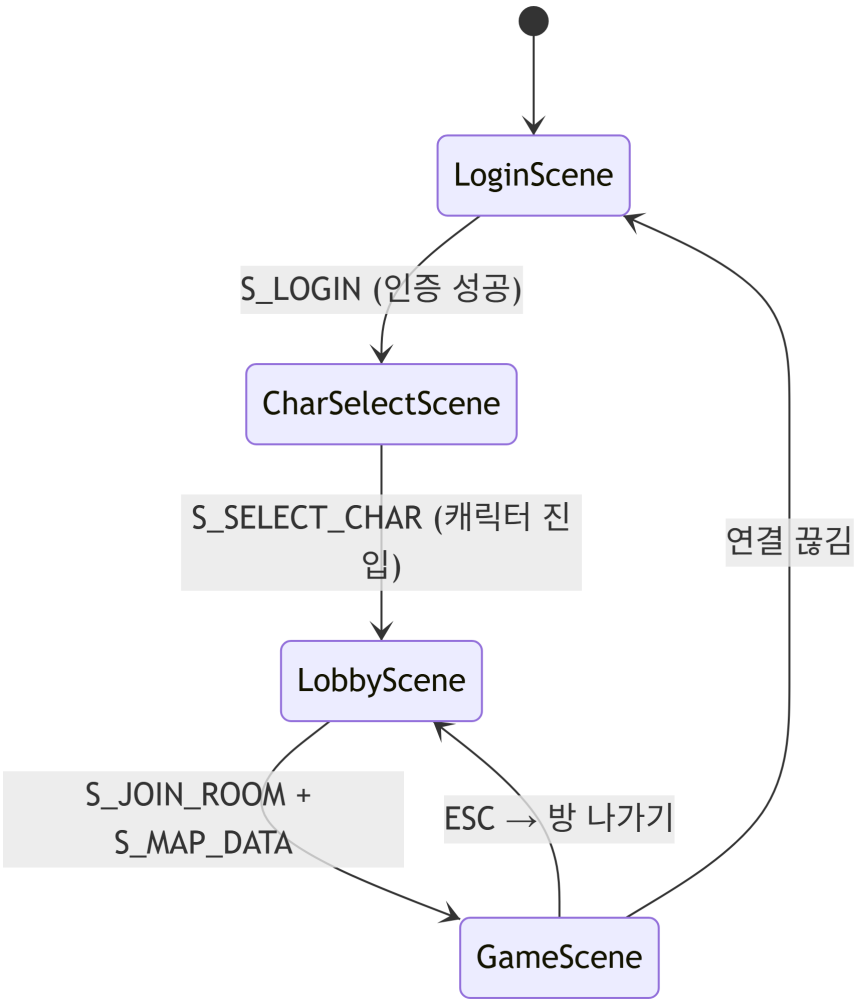
HLSL Constant Buffer 16-byte(float4) . PerFrame `float3 LightDir` `float3 AmbientColor` , GPU `AmbientColor` . `float3` 12 `float3` 16 . `LightDir` `float _pad0` 16 .

Pixel Shader Lambert `HitFlash` , .

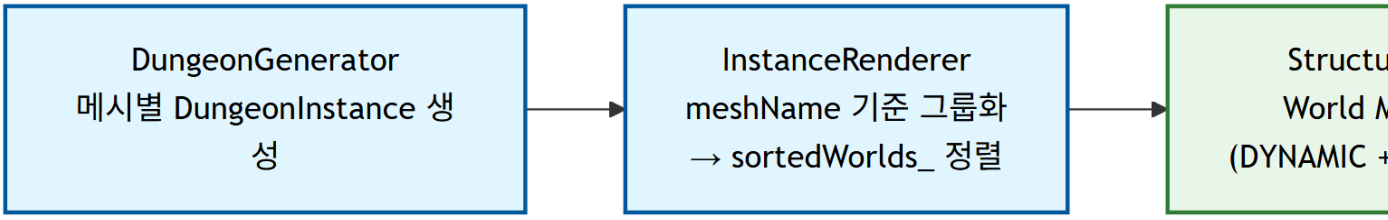
3.2

. `DrawIndexed()` , 300 Draw Call 300 CPU→GPU . floor, wall , Draw Call . `InstanceRenderer` , Structured Buffer World Matrix 1 Draw Call .


DX11Device
(SwapChain·RTV·DSV)
Input
(키·마우스 상태)
UIManager
(ImGui 관리)


LoginScene
S_LOGIN (인증 성공)
CharSelectScene
S_SELECT_CHAR (캐릭터 진입)
LobbyScene
S_JOIN_ROOM + S_MAP_DATA
ESC → 방 나가기
GameScene
연결 끊김

2: 4


DungeonGenerator
메시별 DungeonInstance 생성
InstanceRenderer
meshName 기준 그룹화
→ sortedWorlds_ 정렬
Structu
World M
(DYNAMIC +

3:

4 default.hlsl --- Pixel Shader

```
float4 PSMain(VS_OUT i) : SV_Target
{
    float3 n = normalize(i.WorldNormal);
    float ndotl = saturate(dot(n, -LightDir));
    float4 tex = DiffuseTex.Sample(LinearSampler, i.UV);
    float3 lit = tex.rgb * (AmbientColor + ndotl * 0.7);
    float3 hitColor = float3(1.0, 0.3, 0.2);
    lit = lerp(lit, hitColor, HitFlash);
    return float4(lit, 1.0);
}
```

5:

DungeonInstance	+ World Matrix
BatchInfo	`{startIndex, count}` — Structured Buffer
EnsureBuffer()	(capacity)
BindSingle()	1-element

capacity 1.5 . 256 .

5 InstanceRenderer.cpp --- EnsureBuffer()

```
uint32_t newCapacity = (neededCount < 256) ? 256 : ((neededCount * 3) / 2);

D3D11_BUFFER_DESC bd = {};
bd.ByteWidth = newCapacity * sizeof(XMFLOAT4X4);
bd.Usage = D3D11_USAGE_DYNAMIC;
bd.BindFlags = D3D11_BIND_SHADER_RESOURCE;
bd.MiscFlags = D3D11_RESOURCE_MISC_BUFFER_STRUCTURED;
bd.StructureByteStride = sizeof(XMFLOAT4X4);
```

`InstanceRenderer` Structured Buffer . `instanceBuffer_`(256+ , 1.5) `singleBuffer_`(1-element, Map/Unmap) . non-instanced , . 1-element Structured Buffer .

3.3

. , . 1 , `effect.hlsl` . , .

5 , `EffectRenderer` Right/Up `EffectType` . C++ Slash(1), Bullet(3), Hit(4) 3 , Fireball(0) Heal(2) .

6 effect.hlsl ---

```
float3 worldPos = EffectPos
    + CameraRight * i.Pos.x * EffectScale
    + CameraUp    * i.Pos.y * EffectScale;
```

`EffectTime`(0→1) , .

6:

EffectType			
0	Fireball	+	`smoothstep(0, 0.6·pulse, d)`, `pulse = 1 + 0.15·sin(t·20)`
1	Slash	(arc)	`atan2(y,x)` → `sweepAngle = lerp(-1.5, 1.5, t·3)`
2	Heal		`frac(t + phase)` × 5

7 `effect.hlsl --- Fireball`

```
float4 Fireball(float2 uv, float t)
{
    float d = length(uv - 0.5) * 2.0;
    float pulse = 1.0 + 0.15 * sin(t * 20.0);
    float core = 1.0 - smoothstep(0.0, 0.6 * pulse, d);
    float glow = 1.0 - smoothstep(0.2, 1.0, d);
    float3 innerColor = float3(1.0, 0.9, 0.3);
    float3 outerColor = float3(1.0, 0.3, 0.05);
    float3 col = lerp(outerColor, innerColor, core) * glow;
    float alpha = glow * (1.0 - t);
    return float4(col, alpha);
}
```

8 `effect.hlsl --- Slash ( )`

```
float4 Slash(float2 uv, float t)
{
    float2 centered = uv - 0.5;
    float angle = atan2(centered.y, centered.x);
    float d = length(centered) * 2.0;
    float sweepAngle = lerp(-1.5, 1.5, saturate(t * 3.0));
    float arc = smoothstep(sweepAngle - 0.8, sweepAngle, angle)
              * smoothstep(sweepAngle + 0.1, sweepAngle, angle);
    float ring = smoothstep(0.3, 0.5, d) * smoothstep(1.0, 0.7, d);
    float mask = arc * ring;
    float3 col = lerp(float3(1,1,0.6), float3(1,1,1), mask);
    float alpha = mask * (1.0 - t);
    return float4(col * mask, alpha);
}
```

`EffectRenderer` skillId . Heal (0.5, 1.0, 0.5) 0.6, Slash (1.0, 1.0, 0.6) 0.4, Hit (1.0, 0.3, 0.1) 0.5 .

3.4

`MeshCache` `assets/models/` `.obj` , (Box 0.4×0.9×0.4) (Sphere 0.5) . `ObjLoader` Wavefront OBJ , UV, , .

Kenney 3D OBJ Wavefront OBJ vertex color , `v x y z` . face N-gon(4) , . `ObjLoader` , N-gon face triangle fan . OBJ face normal cross product smooth shading .

`MaterialManager` OBJ MTL `ID3D11ShaderResourceView` , PS `t0` . MTL fallback .

(29): 3 (human, orc, character-a), 3 (floor, floor-detail, wall), 23 (barrel, banner, chest, column, coin, dirt, gate, rocks, stairs, stones, torch, trap, wall-half, wall-narrow, wall-opening, wood-structure, wood-support — Kenney 3D)

4

4.1

(Orthographic) . `Width = orthoSize × aspect × 2`, `Height = orthoSize × 2`, Z [-200, 500] . Pitch 30°, Yaw 45° , (Exponential Smoothing) . Right/Up .

7:

Pitch	30°
Yaw	45°
Distance	20 -
OrthoSize	15 ( ), 5~40 ( )
SmoothTime	0.15s
ZoomSpeed	2.0

.

= + (d·cos(pitch)·sin(yaw), d·sin(pitch), -d·cos(pitch)·cos(yaw))

t = dt / kSmoothTime(0.15s) , pos += (target - pos) * t . SnapTo() .

kZoomSpeed(2.0) orthoSize , [5, 40] / .

4.2 ScreenToGround

. .

(x, y) NDC : ndcX = 2x/w - 1, ndcY = 1 - 2y/h.

NDC near(z=0) far(z=1) invViewProj .

near→far y=groundY t = (groundY - originY) / dirY .

dirY 0 ( ) t = 0 .

9 Camera.cpp --- ScreenToGround()

```
XMFLOAT3 Camera::ScreenToGround(float mouseX, float mouseY,
                                float screenW, float screenH, float groundY) const
{
    float ndcX = 2.0f * mouseX / screenW - 1.0f;
    float ndcY = 1.0f - 2.0f * mouseY / screenH;

    XMMATRIX invVP = XMMatrixInverse(nullptr, GetViewProj());

    XMVECTOR nearPt = XMVector3TransformCoord(
        XMVectorSet(ndcX, ndcY, 0.0f, 1.0f), invVP);
    XMVECTOR farPt  = XMVector3TransformCoord(
        XMVectorSet(ndcX, ndcY, 1.0f, 1.0f), invVP);

    XMVECTOR rayDir = XMVectorSubtract(farPt, nearPt);
    float originY = XMVectorGetY(nearPt);
    float dirY = XMVectorGetY(rayDir);
    float t = (dirY != 0.0f) ? (groundY - originY) / dirY : 0.0f;

    XMVECTOR hit = XMVectorAdd(nearPt, XMVectorScale(rayDir, t));
    XMFLOAT3 result;
    XMStoreFloat3(&result, hit);
    return result;
}
```

4.3

W ' ' (-X, +Z) . Yaw 45° .

screenUp = (-sin45°, 0, cos45°) = (-0.707, 0, 0.707), screenRight = (cos45°, 0, sin45°) = (0.707, 0, 0.707) WASD .

10 PlayerController.cpp ---

```
constexpr float k = 0.70710678f; // 1/sqrt(2)
XMFLOAT3 screenUp    = { -k, 0.0f,  k };  // W:
XMFLOAT3 screenRight = {  k, 0.0f,  k };  // D:
```

8:

W / ↑	(-X, +Z)
S / ↓	(+X, -Z)
A / ←	(-X, -Z)
D / →	(+X, +Z)

Walk 5.0 m/s, Run(Shift) 8.0 m/s , 20Hz(50ms) . 0.0001 0.5° .

11 PlayerController.cpp --- 20Hz

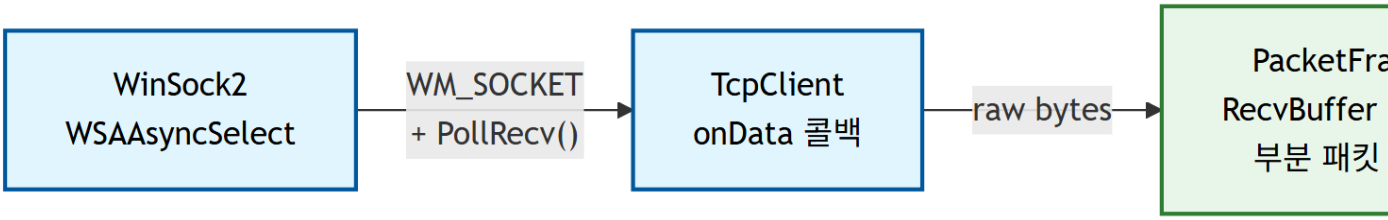
```
sendTimer_ += dt;
if (sendTimer_ >= kSendRate) {    // kSendRate = 0.05s
    sendTimer_ -= kSendRate;
    float distSq = dx*dx + dy*dy + dz*dz;
    float rotDiff = fabsf(data_->rotationY - lastSentRot_);
    if (distSq > 0.0001f || rotDiff > 0.5f) {
        sendPending_ = true;
        lastSentPos_ = data_->position;
        lastSentRot_ = data_->rotationY;
    }
}
```

5

5.1 TCP

, UI Scene . WSAAsyncSelect() PeekMessage/DispatchMessage , I/O . UI .

TcpClient FD_CONNECT, FD_READ, FD_CLOSE WM_SOCKET , PollRecv() FD_READ .



4:

WSAAsyncSelect , Win32 (coalesce) . FD_READ , recv() . PollRecv() non-blocking recv() . WSAEWOULDBLOCK .

12 TcpClient.cpp --- PollRecv()

```
void TcpClient::PollRecv() {
    if (!connected_ || sock_ == INVALID_SOCKET) return;
    uint8_t buf[65536];
    for (;;) {
        int received = recv(sock_, (char*)buf, sizeof(buf), 0);
        if (received > 0) {
            if (onData) onData(buf, received);
        } else {
            if (received == 0) {
                Disconnect();
                if (onDisconnected) onDisconnected();
            }
            break;  // WSAEWOULDBLOCK or error
        }
    }
}
```

5.2

PacketFramer 64KB RecvBuffer TCP . [size:uint16][msgId:uint16][payload] , ProcessBuffer() .

13 PacketFramer.h --- ProcessBuffer()

```
void ProcessBuffer() {
    while (buffer_.ReadableSize() >= 4) {
        const uint8_t* ptr = buffer_.ReadPtr();
        uint16_t size = ptr[0] | (ptr[1] << 8);  // little-endian
        if (size < 4) { buffer_.Clear(); break; }
        if (buffer_.ReadableSize() < size) break;  //    →
        uint16_t msgId = ptr[2] | (ptr[3] << 8);
        if (callback_) callback_(msgId, ptr + 4, size - 4);
        buffer_.Consume(size);
    }
}
```

5.3

§2.3 . NetworkContext .

5.4

59 ID , .

9: ID (59)

101~118 ·	18	Login, Register, EnterGame, CharList, CreateChar, SelectChar, RoomList, CreateRoom, JoinRoom
201~219	19	Move, Spawn, Despawn, Attack, Damage, Fire, PlayerList, SkillData, Respawn, Scoreboard, Portal, GroundItemSpawn/Despawn, Pickup
301~309	9	Chat, CreateParty, JoinParty, LeaveParty, PartyUpdate

501~509	9	InventoryInit, UseItem, DropItem, MoveItem, ItemAdd, ItemRemove
601~604	4	CurrencyInit, Purchase, CurrencyUpdate

6

6.1

, . .

( )

0.4 `ScreenToGround` . 0.6 (y+0.9) `C_FIRE` . , `ProjectileManager` .

(1~4)

`SkillData` `C_ATTACK` . skillId .

10:

skillId	
0	1.0 Slash
1	(Slash)
2	+ C_ATTACK + C_FIRE
3	isSelfTarget → Heal

14 `GameScene.cpp ---`

```
if (skillId == 2) {
    //   :          + C_FIRE
    combatMgr_.HandleFire(app_.playerId, firePos, fireDir, true);
    app_.GetPacketBuilder().SendFire(...);
} else {
    //  :
    combatMgr_.SpawnAttackEffect(fxPos, skillId);
}
```

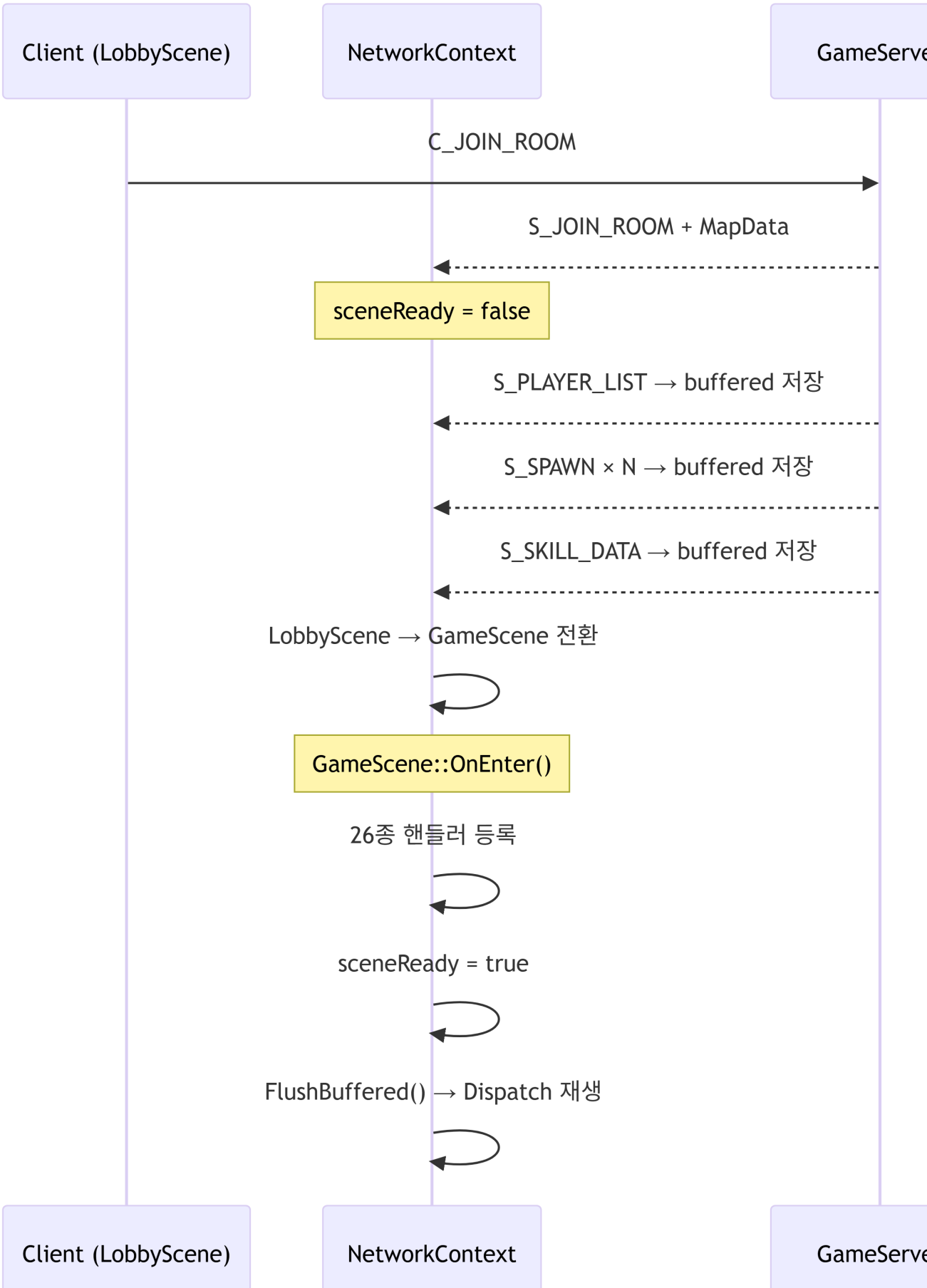
`CombatManager` 4 .

11:

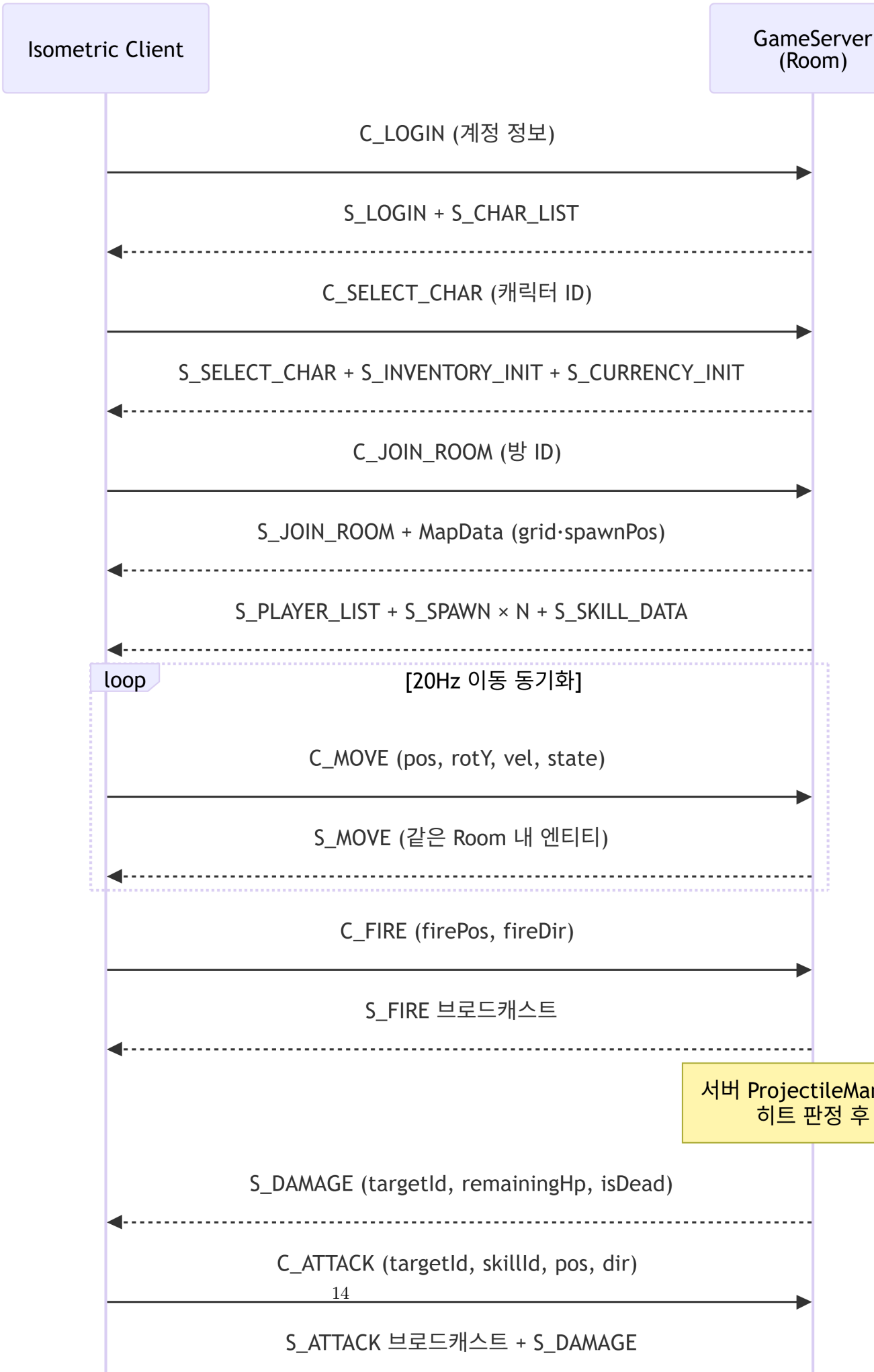
Projectile	80 m/s + `IsWalkable()`	traveled 50m	
AttackEffect	skillId (Slash)	0.3s ( ) / 0.6s ( )	1.0→1.5 ease-out
DamagePopup	ImGui (→)	1.5s	y += 2.0 · dt +
HitEffect	(Hit)	0.3s	y+0.9

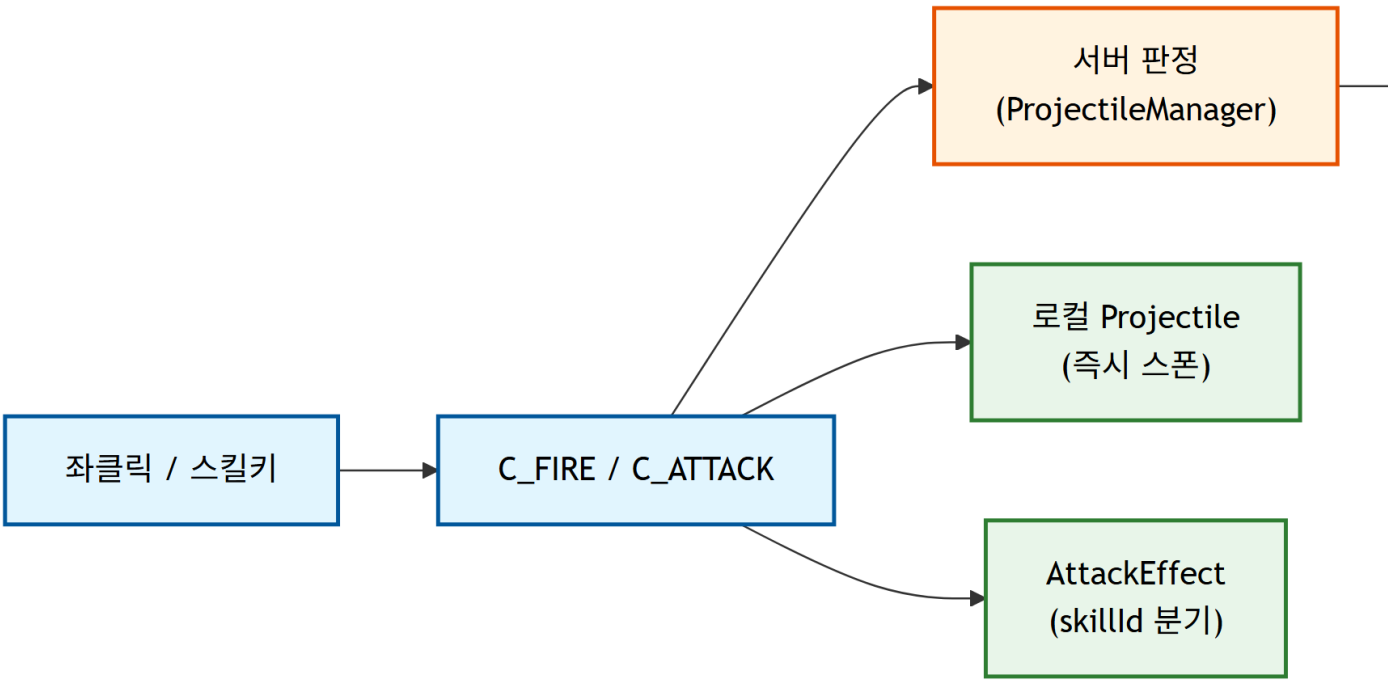
6.2

`DungeonGenerator` `MapData`(grid , cell ,) .


Client (LobbyScene)
NetworkContext
GameServe
C_JOIN_ROOM
S_JOIN_ROOM + MapData
sceneReady = false
S_PLAYER_LIST → buffered 저장
S_SPAWN × N → buffered 저장
S_SKILL_DATA → buffered 저장
LobbyScene → GameScene 전환
GameScene::OnEnter()
26종 핸들러 등록
sceneReady = true
FlushBuffered() → Dispatch 재생
Client (LobbyScene)
NetworkContext
GameServe

5:


Isometric Client
GameServer
(Room)
C_LOGIN (계정 정보)
S_LOGIN + S_CHAR_LIST
C_SELECT_CHAR (캐릭터 ID)
S_SELECT_CHAR + S_INVENTORY_INIT + S_CURRENCY_INIT
C_JOIN_ROOM (방 ID)
S_JOIN_ROOM + MapData (grid·spawnPos)
S_PLAYER_LIST + S_SPAWN × N + S_SKILL_DATA
loop
[20Hz 이동 동기화]
C_MOVE (pos, rotY, vel, state)
S_MOVE (같은 Room 내 엔티티)
C_FIRE (firePos, fireDir)
S_FIRE 브로드캐스트
서버 ProjectileMan
히트 판정 후
S_DAMAGE (targetId, remainingHp, isDead)
C_ATTACK (targetId, skillId, pos, dir)
S_ATTACK 브로드캐스트 + S_DAMAGE



7:

12:

0	floor (10% floor-detail) + 90°
1	,

rand() MapData . , . , MapData .

15 DungeonGenerator.cpp ---

```
auto cellHash = [](int x, int z) -> uint32_t {
    uint32_t h = (uint32_t)(x * 73856093) ^ (uint32_t)(z * 19349663);
    h ^= h >> 16; h *= 0x45d9f3b; h ^= h >> 16;
    return h;
};
// 10% floor-detail: (h % 10) == 0,  : ((h >> 4) % 4) * 90°
```

(1) 4 (N/E/S/W) . , face normal backface culling . (North=0°, East=90°, South=180°, West=270°) . .

16 DungeonGenerator.cpp ---

```
bool adjN = IsFloor(grid, gw, gh, x, z - 1);
bool adjS = IsFloor(grid, gw, gh, x, z + 1);
bool adjE = IsFloor(grid, gw, gh, x + 1, z);
bool adjW = IsFloor(grid, gw, gh, x - 1, z);

if (adjN || adjS || adjE || adjW) {
    float rotY = 0.0f;
    if (adjN)       rotY = 0.0f;
    else if (adjE) rotY = 90.0f;
    else if (adjS) rotY = 180.0f;
    else if (adjW) rotY = 270.0f;
```

17 EntityManager.cpp ---

```
float t = std::min(dt * kLerpSpeed, 1.0f);  // kLerpSpeed = 10.0
for (auto& [id, e] : entities_) {
    e.data.position.x += (e.targetPos.x - e.data.position.x) * t;
    e.data.position.z += (e.targetPos.z - e.data.position.z) * t;

    if (e.hitTimer > 0) e.hitTimer -= dt;

    //     : ±180°
    float diff = e.targetRotY - e.data.rotationY;
    while (diff > 180.0f) diff -= 360.0f;
    while (diff < -180.0f) diff += 360.0f;
    e.data.rotationY += diff * t;
}
```

7

7.1 HitFlash —

, PerObject HitFlash float . 1.0 , .
.

EntityManager hitTimer , GameScene localHitTimer_(0.25) S_DAMAGE .
/ .

7.2 20Hz

60fps 60 . PlayerController 50ms(20Hz) , 0.0001 0.5°
. , .

8

8.1

minimap.hlsl . R8_UNORM , ViewRadius . Vertex Shader SV_VertexID .

( ) ( ) EntityPositions[16] . smoothstep .

8.2 LoL

8 16 → . DungeonGenerator::IsWalkable() , .
.

8.3

.

13:

PlayerController	IsWalkable() +	UX
CombatManager	IsWalkable()	
HandleMove()	GetTile() ,	

. W+D( ) , . PlayerController 3 (Axis-Separated) . X , Z .

8.4 /

PlayerState kills/deaths , Room 5 S_SCOREBOARD . Tab
, .

8.5 +

18 PlayerController.cpp ---

```
float newX = data_->position.x + moveDir.x * speed * dt;
float newZ = data_->position.z + moveDir.z * speed * dt;

if (dungeon_) {
    // 1.
    if (dungeon_->IsWalkable(newX, newZ)) {
        data_->position.x = newX;
        data_->position.z = newZ;
    }
    // 2. X
    else if (dungeon_->IsWalkable(newX, data_->position.z)) {
        data_->position.x = newX;
    }
    // 3. Z
    else if (dungeon_->IsWalkable(data_->position.x, newZ)) {
        data_->position.z = newZ;
    }
    // 4.       -
}
```

8.7 ( )

Room . (R1→R2→R3→R1) .

`GameScene` . column , . Zone depth .

8.8 (M)

M . `ImGui::GetForegroundDrawList()` , 4px .

14:

	(60, 50, 70)
	(140, 130, 150)
+	
+	
(r=4)	
(r=2.5)	
(r=2)	

8.9 UI

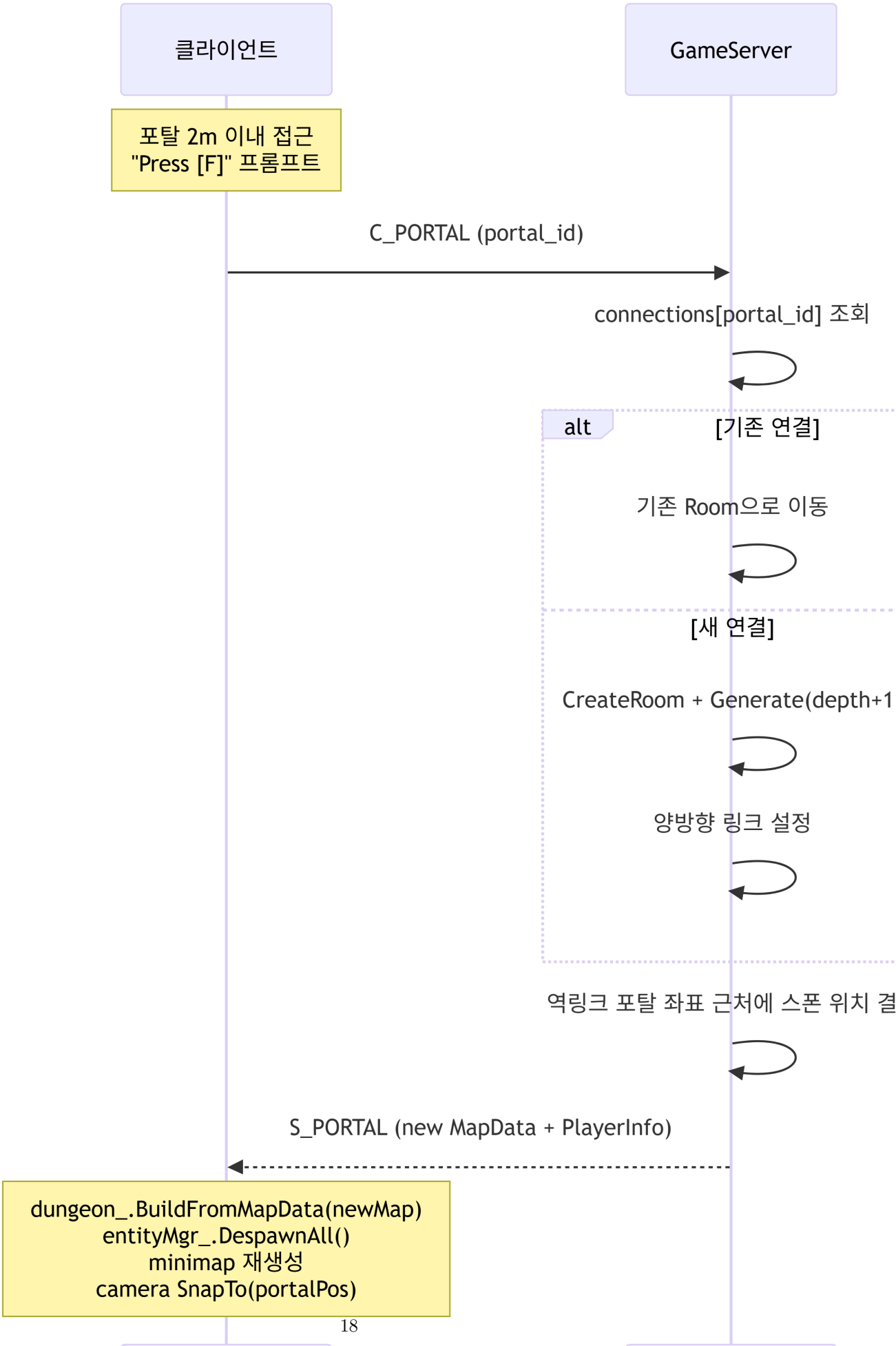
10×10 , `ImGui::Selectable` . (Dog Tag) `GetItemName()` , Use/Drop .

8.10

TCP .

1. /
2. Connect TCP → Login/Register
3. `TcpClient::Connect()` `getaddrinfo()` resolve
4. `lastError_` → UI

Login + .



클라이언트
GameServer

8.11

10m , 3m “Press [E]” . E `C_PICKUP` → → `S_ITEM_ADD` + `S_GROUND_ITEM_DESPAWN` .

coin , 5 . .

8.12 +

10~50 (`S_CURRENCY_UPDATE`) , 30% (`S_ITEM_ADD`) . Gold , HP . +50 Gold .

8.13 GameScene UI

`GameScene::OnUI()` 13 UI .

15: GameScene UI

DrawHPBar		HP · · ,
DrawCrosshair		LoL +
DrawMinimap		
DrawRemoteHPBars	3s	+ HP
DrawDamagePopups	1.5s	
DrawChat		(All/Party/Whisper)
DrawInventory	I	UI +
DrawPartyPanel		HP ·
DrawLeaderboard	Tab	K/D
DrawKillFeed	5s	
DrawFullMap	M	
DrawDeathOverlay		
DrawExitMenu	ESC	

`WndProc` `ImGui::GetIO().WantTextInput` `WantCaptureMouse` UI . , . UP .

8.14 ImGui

ImGui . `ImGui::Begin()` , (I) `Selectable` . ImGui . HP , , , , HUD `ImGuiWindowFlags_NoBringToFrontOnFocus | ImGuiWindowFlags_NoFocusOnAppearing` , .

8.15

16: -

TcpClient + PacketFramer	Session + RecvBuffer	Binary Header + Protobuf
DungeonGenerator.BuildFromMapData()	DungeonGenerator ( )	MapData ·
C_FIRE → → S_DAMAGE	ProjectileManager	+
EntityManager	Room 20Hz	
HandleMove	DungeonGenerator.GetTile()	
S_SCOREBOARD →	PlayerState kills/deaths	/
C_PORTAL →	Room.Connections()	Room
S_GROUND_ITEM_SPAWN → E	Room.SpawnGroundItem()	

S_CURRENCY_UPDATE + S_ITEM_ADD	GrantKillReward()	+ DB
DrawFullMap (M)	MapData +	
getaddrinfo()	Listener 0.0.0.0:7777	

(io_uring , JobQueue , BufferPool,) ServerCore .